AI LAB 99

Machine Learning Track — Week 4, Module 8

Marketing Strategy & Business Dashboard Report

Turning Customer Segments into Personas, Campaigns, Pricing, and a Business Dashboard

Dataset: Final_Customer_Profile_Report.csv (from Module 7)

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Prepared for AI Lab 99 | Customer Segmentation Curriculum

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Introduction & Executive Summary

This module turns the two clusters produced in Module 7 into business-facing deliverables: personas, business-opportunity analysis, marketing strategy, product recommendations, pricing/discount strategy, retention and reactivation plans, a campaign action plan, a merged strategy matrix, and a business dashboard.

Headline finding: of the 12 tasks, 4 are broken and 1 is missing entirely. The root cause in every broken case is the same: this notebook's classification rules (income ≥ 70000, spending ≥ 1000, recency > 60, etc.) were written for raw, real-world-scale data, but the input file's numeric columns are still the StandardScaler output from Module 5/6 — values roughly between −3 and +3. Every threshold check silently fails the same way for both segments, so several "business insight" sections report identical or empty results instead of genuine differences. This is documented task-by-task below.

#	Task	Status	Note
1	**Review Customer Segments**	OK	2 clusters, 1,111 / 927 customers
2	**Develop Customer Personas**	BROKEN	Dollar-scale thresholds on z-score data → both segments "Low"
3	**Analyze Business Opportunities**	BROKEN	Same threshold bug → identical output for both segments
4	**Marketing Strategy Design**	OK	Name-based logic — correctly differentiated
5	**Product Recommendation Strategy**	OK	Name-based logic — correctly differentiated
6	**Pricing & Discount Strategy**	OK	Name-based logic — correctly differentiated
7	**Customer Retention Strategy**	OK	Name-based logic — correctly differentiated
8	**Customer Reactivation Strategy**	BROKEN	Recency > 60 on z-score data → 0 inactive customers found
9	**Campaign Action Plan**	OK	Name-based logic — correctly differentiated
10	**Marketing Strategy Matrix**	BROKEN	Inner join with empty Task 8 output → entire matrix is empty
11	**Business Dashboard**	PARTIAL	Charts render, but "Total Revenue" / "Avg Income" are z-score sums, not currency
12	**Management Summary Report**	MISSING	Section header only — notebook ends with no content

Task 1 — Review Customer Segments

The Module 7 output (2 clusters, 2,038 customers) was reloaded and confirmed: Cluster 0 = 1,111 customers (54.51%), Cluster 1 = 927 customers (45.49%). Business names were assigned from a 4-entry dictionary (Premium Loyal Customers, Budget Buyers, At-Risk Customers, New Customers), of which only the first two map to an actual cluster — consistent with the segment-count mismatch already flagged in the Module 7 report.

Naming inconsistency across modules: Module 7 named Cluster 1 “Digital-First Buyers”; this notebook renames the same cluster “Budget Buyers.” The underlying data didn't change — only the label did — so downstream materials from the two modules will describe the same customers with two different names. Standardize on one naming convention before these documents reach stakeholders.

Task 2 — Develop Customer Personas

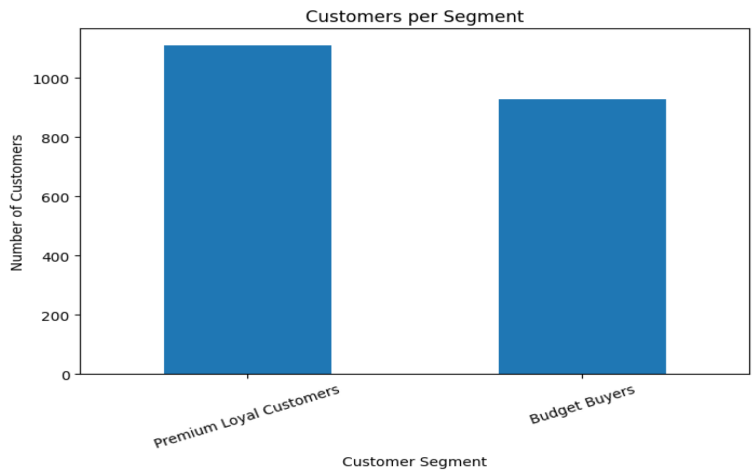

Figure 1. The two segments really are very different in scale — which makes the identical persona labels below all the more clearly wrong.

This task's rule-based classifier is broken. As delivered, both segments are labeled identically:

Premium Loyal Customers (Income +0.67, Spending +0.81)	Income Level: Low · Spending: Low · Purchase Frequency: Rare · Campaign Response: Low · Lifetime Value: Low
Budget Buyers (Income −0.80, Spending −0.97)	Income Level: Low · Spending: Low · Purchase Frequency: Rare · Campaign Response: Low · Lifetime Value: Low

Every threshold in the classifier (income ≥ 70000/40000, spending ≥ 1000/500, purchases ≥ 15/8, campaign ≥ 3/1) is calibrated for raw-scale data. Because both segments' z-scores fall far below every one of these cutoffs, both fall into the lowest bucket on every dimension — despite Premium Loyal Customers being the objectively higher-income, higher-spending group by a wide margin. Only Deal Dependency (thresholds 0.60/0.30, which happen to sit inside the z-score range) and the raw Customer_Activity_Level passthrough actually differentiate the two personas as delivered.

Fix: either inverse-transform Income/Total_spending/Total_Purchase back to their original scale before applying these thresholds, or rewrite the thresholds in z-score terms (e.g., income ≥ 0.5 → “High Income”) so they're calibrated to the data they actually receive.

Task 3 — Analyze Business Opportunities

Same bug, same result: the business-opportunity generator uses the same raw-scale thresholds (spending ≥ 1000/500, campaign ≥ 2, recency ≥ 60) and produces word-for-word identical output for both segments

— “Large customer base / Opportunity for future growth,” “Increase campaign engagement,” “Moderate risk of switching competitors,” and “Convert low-value customers into active buyers” — even though Premium Loyal Customers is the highest-value segment in the dataset. This section currently gives zero differentiated business guidance and should not be used until the same unit fix from Task 2 is applied here.

Task 4 — Marketing Strategy Design

This task and the four that follow it (5–7, 9) use name-based string matching ("Premium" in segment, "Budget" in segment) rather than numeric thresholds, so they correctly differentiate the two segments regardless of the underlying scaling issue.

Premium Loyal Customers	“Exclusive Premium Collection for Our Best Customers” · Professional & Exclusive tone · VIP Membership & Premium Discounts · Email · Weekend Evening · Weekly
Budget Buyers	“Save More with Special Discounts” · Friendly & Promotional tone · Coupons & Cashback Offers · SMS · Month End · Bi-Weekly

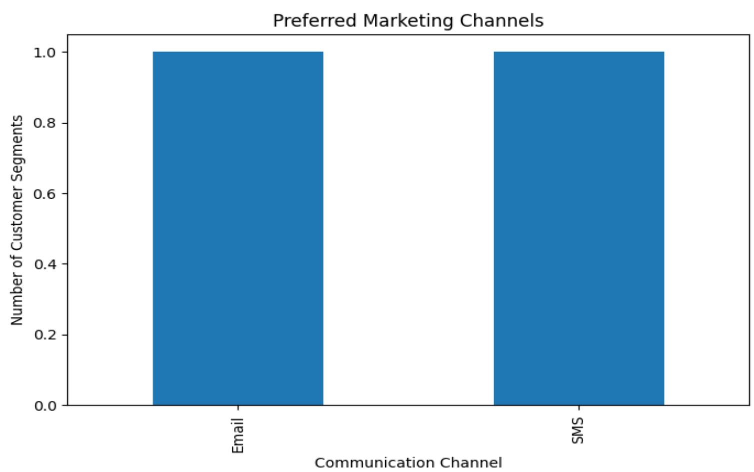

Figure 2. Preferred marketing channel by segment.

Task 5 — Product Recommendation Strategy

Premium Loyal Customers	Primary: Wine · Secondary: Gold Products · Cross-sell: Premium Meat · Upsell: Luxury Gift Hampers · Bundle: Wine + Gold Products
Budget Buyers	Primary: Sweets · Secondary: Fruits · Cross-sell: Fish Products · Upsell: Family Value Packs · Bundle: Fruits + Sweets Combo

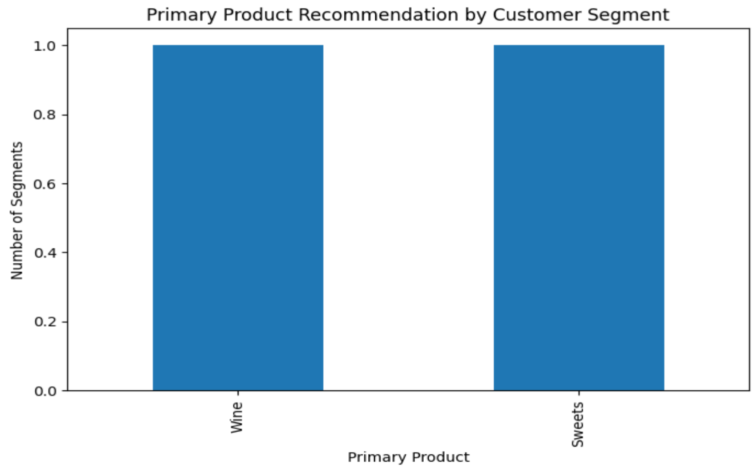

Figure 3. Primary product recommendation by segment.

Task 6 — Pricing and Discount Strategy

Premium Loyal Customers	5% discount · Exclusive VIP Coupons · Luxury Gift with Purchase · Premium Holiday Collection · Premium pricing: Yes
Budget Buyers	25% discount · Weekly Discount Coupons · Buy 2 Get 1 Free · Festival Discount Sale · Premium pricing: No

Figure 4. Recommended discount percentage by segment.

Task 7 — Customer Retention Strategy

Premium Loyal Customers	VIP Reward Points · Exclusive Loyalty Program · Weekly follow-up · High support priority · Gold Membership · Premium Referral Bonus
Budget Buyers	Cashback Rewards · Discount Campaign · Bi-weekly follow-up · Medium support priority · Silver Membership · Discount Coupon

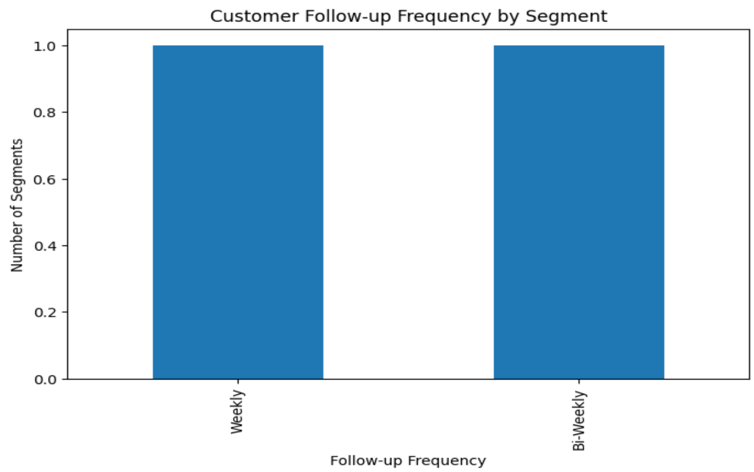

Figure 5. Follow-up frequency by segment.

Task 8 — Customer Reactivation Strategy

This task's output is completely empty. Inactive customers were identified with df[df["Recency"] > 60] — a raw-days threshold applied to a standardized Recency column whose values range roughly from −1.7 to +1.7. No row can ever exceed 60, so the inactive-customer filter returns 0 rows, the per-segment inactive summary is empty, and the entire reactivation plan table that depends on it is built with zero rows. As written, this task delivers no reactivation plan at all.

Fix: inverse-transform Recency back to days before filtering, or use a z-score-appropriate cutoff (e.g., Recency > 1.0 for "above-average days since last purchase").

Task 9 — Campaign Action Plan

Like Tasks 4–7, this task uses name-based logic and works correctly:

Premium Loyal Customers	Campaign: VIP Rewards · Objective: Increase Customer Loyalty · Email · Budget: High · KPI: Repeat Purchase Rate
Budget Buyers	Campaign: Discount Week · Objective: Increase Sales · SMS · Budget: Medium · KPI: Conversion Rate

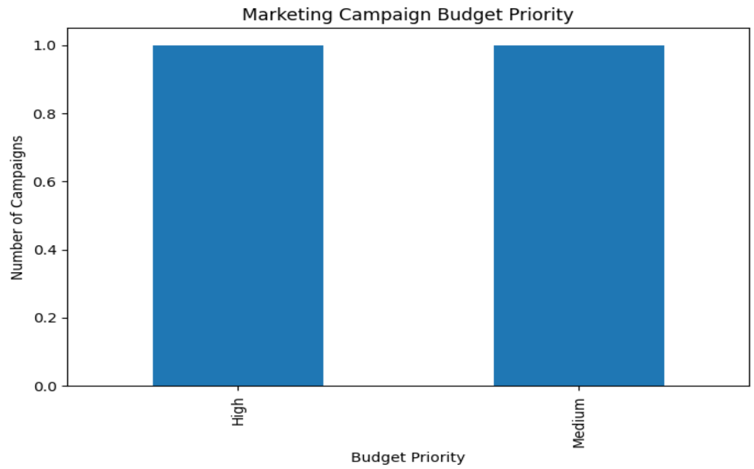

Figure 6. Campaign budget priority by segment.

Task 10 — Marketing Strategy Matrix

This task cascades the Task 8 failure into total data loss. The final matrix is built by inner-joining the marketing, product, discount, retention, and reactivation tables on "Customer Segment." Because the reactivation_plan table from Task 8 has zero rows, the inner join drops every row from every other table too — even though Tasks 4–7 and 9 each independently produced valid, correctly-differentiated 2-row tables. The final_strategy_matrix that this task is meant to deliver has 0 rows and 10 empty columns.

Fix: fix Task 8 first, or use a left join anchored on the full segment list instead of a chain of inner joins so one empty table can't silently erase the rest.

Task 11 — Business Dashboard

The six dashboard charts all render successfully and are visually correct given their inputs, but two of the headline summary numbers are not meaningful business metrics as printed:

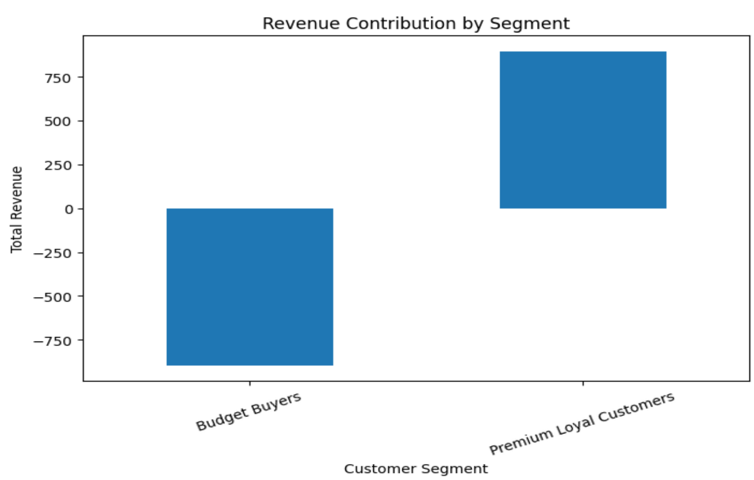

Figure 7. "Revenue Contribution by Segment" — Budget Buyers shows as roughly −900, a negative total that cannot be a real revenue figure.

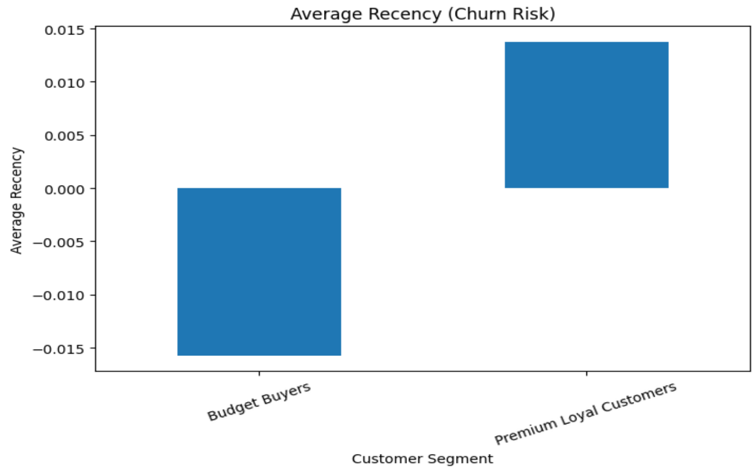

Figure 8. "Churn Risk by Segment" (average Recency) — both segments sit within 0.02 of zero, showing essentially no differentiation once plotted at this scale.

Total Revenue (dashboard summary)	−1.9 — a sum of standardized spending values, not a currency figure
Average Customer Income (dashboard summary)	0.0 — the mean of a standardized column is by definition ≈ 0
Average Spending (dashboard summary)	−0.0 — same issue

Fix: either inverse-transform these three columns before the dashboard summary is computed, or drop them from the printed summary and keep only the correctly-scaled comparison charts, which remain useful for relative (not absolute) reading.

Task 12 — Management Summary Report

Not implemented. The notebook contains only the "# Task 12: Management Summary Report" header cell; no code or narrative follows it. This deliverable is currently missing and should be authored as a follow-up — ideally after Tasks 2, 3, 8, and 10 are fixed, since a management summary built on top of the currently broken sections would repeat their errors (identical low-tier personas for both segments, an empty reactivation plan, an empty strategy matrix, and a negative revenue figure).

Limitations & Recommendations

- **Root-cause fix first:** the single highest-leverage fix is inverse-transforming Income, Total_spending, Total_Purchase, and Recency back to their original units (or an unscaled copy of the dataframe) before any threshold-based rule runs. This alone would repair Tasks 2, 3, 8, 10, and the Task 11 summary numbers.
- **Prefer robust joins:** Task 10's chained inner-joins let a single empty table (Task 8) silently zero out four other tasks' valid work. A left join anchored on the segment list, with a data-quality assertion on row counts, would have caught this immediately instead of shipping an empty matrix.

- **Reconcile segment names across modules:** Cluster 1 is “Digital-First Buyers” in Module 7 and “Budget Buyers” in this module. Pick one name and use it everywhere downstream.
- **Only 2 of 4 named segments exist:** as in Module 7, this notebook's marketing/discount/retention/campaign tables are only ever populated for 2 segments even though logic exists for 4 (“New” and “Risk” branches are unreachable with the current clustering). Either re-cluster at a higher k or remove the unreachable branches to avoid maintaining dead code.
- **Add sanity checks:** a simple assert len(df) > 0 after each filter/merge step would have caught the Task 8 and Task 10 failures immediately instead of letting an empty dataframe propagate silently to the final deliverable.

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